Aaron Chu

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Education

Rutgers University — New Brunswick, NJ

BS Computer Engineering — *May 2028*

- **Coursework**: Software Engineering, Data Structures, **Distributed Deep Learning**, Computer Architecture, Discrete Mathematics, Probability & Random Processes

Experience

Software Engineering Intern — June 2026 – Aug. 2026

Magnefy — *Palo Alto, CA*

- Automated processing of sensor data from 9 industrial transformers (598 captures, 193K waveforms) with an **S3→SQS** pipeline on 4 **Railway** services, replacing a manual process; operators approve results before publishing
- Replaced an **ML** anomaly filter with a rule-based screen on raw samples that caught all 695 labeled arcing events at a 1.48% false-positive rate, now running in production
- Built the production **PostgreSQL** persistence modules and schema (121 of its 140 columns), and **AWS SES** email alerts with stable IDs so reprocessing never sends duplicates
- Trained an **XGBoost** classifier that tells 3 partial-discharge types apart (0.991 macro F1)

Lead Systems Integration Engineer — Sept. 2025 – Present

University Nanosatellite Program — *New Brunswick, NJ*

- Led a 10-member team and built a thread-safe **Python** package polling 4 **I2C** sensors with concurrent workers and serialized bus access on a **Raspberry Pi**, validated by 14 hardware-free tests
- Prevented stale-data bugs by tracking per-sensor errors and reading age with a monotonic clock, filtering invalid readings, and regression-testing failure and recovery
- Authored 3 Critical Design Review test procedures (communications, attitude control, thermal) for SPICEsat, a 6-unit **CubeSat**, tracing 20+ requirements to verification methods

Drone Support Intern — June 2025 – Aug. 2025

Integem — *Cupertino, CA*

- Built high-altitude balloon telemetry: **Arduino** C++ firmware sent GPS and pressure data over **LoRaWAN/Helium** with geofenced switching across 5 radio regions, decoded into **InfluxDB/Grafana** for a live map
- Implemented 3 **PID** loops (altitude, forward, yaw) on an **ESP32-S3** drone and fused **AprilTag** camera headings with the IMU to cancel yaw drift for autonomous flight, tuned from logged flight data

Projects

DistributeDB | *solo project* — github.com/aarochu/DistributeDB

- Built a crash-durable, replicated key-value database from scratch in **Rust** (about 19K lines, standard library only): checksummed write-ahead log, group commit, and an **LSM-tree** storage engine
- Added primary-to-replica log replication with disk-durable acknowledgments and optional semi-synchronous mode
- Killed the primary during 100K concurrent writes and verified every acknowledged write survived restart; cut 99th-percentile read-lock wait by 82 to 89% by moving disk syncs outside the lock

CUDATracker — github.com/aarochu/CUDATracker

- Built a real-time object detection and **SORT** tracking pipeline in **C++/Python** with custom **CUDA** preprocessing kernels validated against CPU output
- Benchmarked 9 configurations of **PyTorch**, **ONNX**, and **TensorRT** with CPU and CUDA preprocessing; FP16 TensorRT with CUDA reached 77.1 FPS on a laptop GPU, but CUDA did not help PyTorch FP32

DeskPartner — github.com/aarochu/DeskPartner

- Architected the 5-module stack (arm control, vision calibration, **VLM** planner, data collection, training) for a 5-person team; earned Honorable Mention at the Embodied Metal Hackathon
- Ported **Rerun**'s robotics logging toolkit (record, catalog, query, replay) to an unsupported 7-joint arm, open-sourced it, and led the team's Rerun bounty integration

Technical Skills

Languages: Python, Rust, C++, C, Java, TypeScript/JavaScript, SQL
Systems and Backend: AWS, PostgreSQL, Docker, Linux, FastAPI, Node.js, Git, pytest
ML and GPU: CUDA, PyTorch, TensorRT, ONNX, XGBoost